

13. Other Avenues to FCL

V 2.5

Please note that with autoISF you are in an early-dev. environment, where the user interface is **not optimized for safety** of users who stray away from intended ways to use. Good safety features exist, but these are only as good as the development-oriented user understands and implements them. This is not a medical product, refer to disclaimer in [section 0](#)



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Available related case studies:

Case study 13.1: Comparison 1 month FCL Automation vs autoISF
Case study 13.2: FCL using dynamicISF (call for an example so far un-answered)
Case study 13.3: FCL using Boost

13.1 Full Closed Loop using AAPS Master and Automations

AndroidAPS 3.0 was (Sep.2023) the first DIY system to launch Full Closed Looping as an option to manage T1D, if a described set of pre-requisites apply.

Key pre-requisites were described in

<https://androidaps.readthedocs.io/en/latest/Usage/FullClosedLoop.html> , and are sketched also in [section 1](#), with [case studies 1.1 – 1.5](#) underscoring the importance.

You may (not) have noticed: There was no big „marketing fuzz“ made around that FCL option. Seeing how many AAPS users struggle with even getting their basal, ISF and SMB settings right, it would be foolish to allure everybody to a supposedly very easy way of looping. True, it can be easy. But only after doing a personalized set-up project. Setting up is easier than what autoISF and the methods we get to in [section 13.3](#). demand, but still a project. It also requires a well mastered hybrid closed loop, to start from.

With attention to the pre-requisites, and avoiding extreme high carb diets, many (mostly: adult) users achieve satisfactory %TIR after supplementing AAPS Master with personalized Automations that attempt to strongly elevate iob upon recognition of a meal-related bg rise.

43 See also Case Studies, and the randomized cross-over study involving AAPS FCL: PubMed [First](#)
44 [Use of Open-Source Automated Insulin Delivery AndroidAPS in Full Closed-Loop Scenario:](#)
45 [Pancreas4ALL Randomized Pilot Study](#);

46

47 This method is **highly recommended for an entry into FCL for those who do not have the**
48 **interest, or lack the time, to deal with the very much more sophisticated and demanding other**
49 **routes** towards FCL, like autoISF, or also like the methods briefly presented below in [section 13.3](#).

50

51 Note that using the autoISF dev version of AAPS for this (*with "Enable ISF adaptation.." OFF*)
52 can be a good idea, to make use of features like SMB_range_extention and
53 SMB_delivery_ratios > 0.5. Compared to using AAPS Master, this allows stronger boosting of
54 SMB sizes, also when *not making use of autoISF*, but just of Automations, for FCL.

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56 13.2 FCL using dynamicISF with AAPS or with Trio / iAPS

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58 As opposed to

- 59 • autoISF, with it's bgAccel_ISF component , or to....
- 60 • AAPS Master, with Automations strengthening ISF at meal-related bg rises ...

61 dynamicISF was **not** designed to help boost SMBs asap after an omitted user bolus.

62 This is why methods like Boost ([section 13.3.1](#)) that even do have dynamicISF on bord, do extra
63 boosting routines for detected or announced meal starts.

64 Rather (as the name also suggests) it was designed to be used in hybrid closed looping to make
65 ISF react more dynamic to suspected swings in insulin sensitivity (which shows in bg values, and
66 in TDD trends). It does a similar job like Autosens, but can be much more amplified (by the users
67 tuning their dynamicISF adjustment factor (%)).

68

69 When using a fast insulin (and when some other pre-requisites discussed in section 1 are in place,
70 too), the dynamicISF method can be applied also to Full Closed Looping. (See [case study 13.2](#);
71 *not available by time of publication => This is a call for a dynISF FCL user to provide a case study*
72 *that contains a 1 week 24h scatter plot as well as one analyzed meal where we can see when and*
73 *how dynISF helped build iob, after not having bolussed*).

74

75 It will have a principal timing-disadvantage because responses are more tied to high bg values
76 than to acceleration (in autoISF) or to delta (in the Automations route to FCL).

77

78 On the other hand, people who 1) do have strong sensitivity swings and 2) cannot pro-actively
79 deal with those (e.g. by making profile switches) might be satisfied with the automatic (although a
80 bit late) adjustments that dynamicISF automatically will provide.

81

82 dynamicISF therefore could be characterized, in the FCL context, as a potential solution to a rather
83 care-free approach for those who do not seek best-possible performance (or who take other
84 measures, like low carb diet, to still reach pretty acceptable performance in FCL mode).

85

86 **More info** (caution, both not focussed on FCL:)

87 AAPS / search term dynamicISF in: <https://discord.gg/DfvK5HnxXu>

88 Trio or iAPS / section dynamic-isf-cr: <https://discord.gg/gKXW5uX3m>

89

90

91 **13.3 Methods involving simple Meal Announcement that might be stretched** 92 **into a Full Closed Loop**

93

94 See also [section 7](#) on using autoISF in “MA” mode, involving a pre-meal bolus.

95

96 **13.3.1 Boost**

97

98 All of the additional code outside of the standard SMB calculation requires a daily time period
99 („Boost window“) to be specified within which it is active.

100 A variation of dynamicISF is used in which also predicted bg will be considered in varying degrees
101 (40...75%) to mimic the effects of higher insulin sensitivity at lower glucose levels.

102 When using Boost without carb inputs (permanent cob=0) a special **boosting of SMBs** is provided
103 when an **initial bg rise** is detected with a meal:

104 Boost uses the delta accelerations to drive an expectation of a higher future levels, allowing more
105 insulin to be delivered, so it principally can also work with no meal announcement (see [case study](#)
106 [13.3](#)). delta, short_avgDelta and long_avgDelta are used to trigger an early bolus (assuming IOB is
107 below a user defined amount).

108 This procedure goes in the direction of the bgAccel_ISF route discussed for autoISF
109 ([section 4.1](#)). If used with an excellent CGM, autoISF acceleration detection should be a bit
110 earlier, and boosting can be made much stronger in autoISF

111 For safety, the user sets a value of 2.5% (up to 5%) of TDD for the max. Boost Bolus (Boost Bolus
112 Cap).

113 For stronger boost, the default AAPS 50% SMB_delivery_rate can be overwritten with a higher in-
114 sulin percentage determined by the user. The SMB_delivery_ratio is called „Boost insulin required

115 percent“ here, and suggested not to go over 75%. The % can be defined variable with bg value
116 (like also in autoISF).

117 The Boost function automatically shuts off as soon as delta and the average deltas are aligned,
118 i.e. when the accelerated rise goes over into a constant rise (compare pp_ISF in autoISF).

119 However, the boost function is only „dormant“ if the boost window lasts longer for more meal-
120 related accelerations.

121 Additional functions are a step-count modified dynamic_ISF, inactivity detection etc

122 A couple of safety feature are integrated. The user can define an iob limit for boosts (like iobTH in
123 autoISF, here called UAM Boost max IOB in Preferences/Treatments) There is also a user adjusta-
124 ble Low Glucose Suspend threshold. This allows the user to set a value higher than the system
125 would normally use, such that when predictions drop below this level (65...100), a zero TBR is set.

126 Application example, see [case study 13.3](#)

127 More info: <https://discord.gg/nYC4T9PgCR> ; <https://github.com/tim2000s/no-bolus-dev>
128 : https://github.com/tim2000s/Boost_AAPS_3.2/blob/Boost-Master-3.2/README.md

129 Contact: Tim Street @ diabettech.com

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131 **13.3.2 AIMI** (available on AAPS and on Trio platform)

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133 AIMI has a single goal: to minimize the decisions necessary to maintain the target range, simplify
134 the composition of the profile for the user or doctor accompanying the patient, and allow the patient
135 to live normally without having to count carbohydrates or even without signifying physical activity
136 (especially for brisk walking).

137 A key component of AIMI concept is to give a **small pre-bolus before each meal** (“Meal
138 Announcement” that also provides some pos. iob).

139 • A **simplified profile** composition (neutral ISF around 100, DIA 9, target 90-90, a single value for
140 basal, a ratio that is not used in AIMI, so not important) For a first basal estimate, you can use the
141 TDD / weight ratio.

142 • Some variables in preferences that are important (AIMI_UAM which allows AIMI to make
143 decisions, Max SMB size which is the highest value for an SMB, B30_duration (which is the
144 duration during which the **basal will be forced after a manual bolus**), B30_upperBG and
145 B30_Upperdelta (these last two variables represent the conditions for replacing smb with a
146 consistent TBR depending on the delta)

147 • The basal profile is calculated by a polynomial equation.

148 • The ISF is calculated from the TDD (**dynamicISF**) and is adjusted based on the evolution of TIR
149 throughout the day and the **detection of physical activity**.

150 • The detection of glycemic rise (or the opposite situation) is also calculated by a polynomial
151 equation, which will influence the change of target but also the replacement of SMB by a TBR
152 between 100% and 500% or by an SMB of the same equivalence.

153 • SMB calculation is done in several ways specific to AIMI depending on the evolution of the delta
154 and IOB, with a distribution that can be done in three parts depending on the conditions.

155

156 Example scenario of execution, on almost all existing variants:

- 157 1. Make a "standard" manual bolus. I usually do 1.5U or 2U with luymjev
- 158 2. Just after this bolus, AIMI will force the 500% TBR for a duration defined by the user. The
159 observation made is that the absorption of insulin such as humalog for example is acceler-
160 ated and will strongly limit the first wave.
- 161 3. Depending on the options chosen, it is possible to receive an SMB of the initial manual bo-
162 lus size after the duration of the 500% TBR
- 163 4. Then the rest of the calculations will depend on the result of a polynomial equation and its
164 evolution.
- 165 5. A few hours later, if the patient decides to take a walk to go shopping, or other activities re-
166 quiring movement, the phone sensor will send information on the number **of steps taken**.
167 This will result in a reduction of the profile to about 60%. The return of the profile to normal
168 will be done in stages, in the first half hour following the activity, the profile will be restored
169 to about 80%.

170 The AIMI developer has been working on incorporating machine learning (using tensorflow lite).

171 More info <https://discord.gg/jnDnJtcBb7>

172

173 The developer hasn't kept the code public. AIMI can only be obtained as an apk via joining their
174 WhatsApp group or here:

175 https://github.com/MTR93600/OpenApsAIMI/tree/dev_mergemilos_addOAPSAIMI

176

177 Given the very high number of changes happening in this AAPS variant, it is probably deemed
178 important to keep it in a tight sub-community. But, caution: This can be seen as violation of the
179 Open Source principle

180

181 Contact: Mathieu Tellier @ AndroidAPS User FB / Twitter @MTR93600/, Discord: MTR

182

183 13.3.3 EatingNow (EN)

184

185 This version of AAPS has evolved over time using elements from AIMI and Boost. It includes a
186 modified dynamicISF which moves ISF modulation in the direction as pioneered by autoISF, and
187 also uses Automations for FCL.

188 "Eating Now" (EN) allows user definable SMB's when deltas are sufficient and accelerating.

189 The intent of this plugin is the same, to deliver insulin earlier using mostly the AAPS predictions.

190

191 As all other variants for FCL, also EatingNow requires to set glucose TT occasionally, to nudge the
192 loop in certain direction, notably to announce and be prepared for exercise.

193

194 Operating Modes provide 3 levels of „aggressiveness“ in 3 time windows:

- 195 • Master AAPS w/up to 120 min basal per SMB when EN is off (usually set for night-time).
- 196 • EN (usually set for daytime) is when the modified algorithm is capable of boosting ISF and
197 insulin delivery. At BG level rises within the EN Window, a „UAM maxBolus“ is given as a
198 first SMB. Recommended Setting: 1h current basal in units (max allowed: 2).
- 199 • ENW: A further boosted SMB will be issued in this ENW time window (e.g. for breakfast, or
200 generally for the first meal of a day, after fasting, with higher insulin need). Upon detection
201 of rising glucose, a SMB called Breakfast COB maxBolus is given by the loop. Recom-
202 mended Setting: 25% of average breakfast total units

203

204 EN uses the dynamicISF concept, modified to making ISF stronger with increased eventualBG
205 predictions.

206

207 Specifically for the ENW (usually: breakfast window), an additional boost factor called Breakfast
208 ISF/CR Percentage (e.g. 125 or 150%) can be applied

209 A setting „TIRS“ provides a very simple version of autoISF (dura_ISF) and sharpens ISF
210 temporarily when bg „seems stuck“ above a certain value.

211

212 Autosens sensitivityRatio will be overridden by EN sensitivity options.

213

214 SMB delivery ratio for insulinReq. is set to 65% for when EN is disabled (overnight, usually).

215 It is recommended to set maxSMBBasalMinutes and maxUAMSMBBasalMinutes to 30 minutes
216 max as these will be used when EN is OFF or in SLEEP mode. Falling back on OpenAPS SMB
217 settings is considered as the safe mode, should you experience any issues with sensitivity or EN
218 settings in general

219 It is set 85% for an active ENW, or 75% when EN is on but ENW not active

220

221 Furthermore, SMB optionally can be disabled day/night below defined bg level/s (SMB Disabled)

222 **More info** <https://discord.gg/XqhnPRChEP> (method description in pinned post)

223 <https://github.com/dicko72/AAPS-EatingNow> scroll down to README.md

224 Contact: dicko via Discord channel

225

226 **13.3.4 Tsunami**

227

228 The Tsunami loop algorithm analyses blood glucose and insulin activity developments to estimate
229 bolus requirements during meals, without the necessity of carb announcements.

230

231 Users must make a **meal announcement via a button** on AAPS main screen. It switches on the
232 main Tsunami algorithm for a finite amount of time.

233

234 In between meals (when Tsunami is inactive), users are given the choice between running a
235 weaker version of the Tsunami algorithm (called wave), or falling back tooref1.

236

237 A “historic” merit of this method was that it pioneered a BG smoothing algorithm that later
238 became included as a plugin in AAPS.

239 The insulin models dynamically readjust DIA based on bolus size so that a user-set, fixed
240 DIA value is no longer needed.

241

242 For best results, it is recommended to issue a **bolus** at the beginning of a meal to account for the
243 disadvantageous kinetics of subcutaneously administered insulin in a UAM setting.

244 **More info** <https://discord.gg/veRKcgwVUT> GitHub repository: <https://github.com/piecycle/tsunami>
245 [official documentation: https://cdn.discordapp.com/attach-](https://cdn.discordapp.com/attachments/969948954949189633/972852790739238992/tsunami_guide_3_2.pdf)
246 [ments/969948954949189633/972852790739238992/tsunami_guide_3_2.pdf](https://cdn.discordapp.com/attachments/969948954949189633/972852790739238992/tsunami_guide_3_2.pdf)

247

248 Contact: nichi#1391 on discord / piecycle on GitHub

249

250 **13.4 No-Bolus Looping with Carb Entries**

251

252 Some oreof(1) loopers attempting to go full closed loop reported that they do best when they (do not
253 bolus but) give their loop precise carb (and absorption time) information. This:

254 * announces a meal to follow (so it is not UAM, but might be called full closed looping if the
255 insulin management is left 100% to the loop)

- * provides data on cob, and with the glucose and insulin activity info the loop has, it can always calculate how much more carbs are to become absorbed (to the extent the carb-related infos the user put in is correct)
- * will display realistic cob info to the user, including cob info looking forward (rather than only calculating carb deviations for the past minutes or hours, and making some coarse assumptions for the upcoming hour). It gives the user better feeling of safety if she/he can see cob info in addition to the available iob info, and insulin activity prediction.

With detailed carb (amounts + absorption times) inputs, the loop has best-possible info to provide „the best expert fit“ of insulin activity and carb absorption.

It still rarely can come close to physiological values, because the time-delays inherent in our „artificial pancreas“, notably the stretched out DIA, make it difficult still, compared to a real pancreas.

So, carb inputs could help. However,

- only to the extent amounts and time pattern for absorption („eCarbs“) are correct ((which, every day, is pretty much a mission impossible))
- the oref(1) loop still largely „waits for glucose to rise“, and there is no significant time advantage from inputting carb info

Only the **user**-bolussing *for expected* carb absorption in hybrid closed loop offers a convincing time advantage (but with associated risks).

- inputs require actually more attention to detail than it is good practice even in AndroidAPS hybrid closed loop, so in that respect a step back, not forward.

Entering **precise** carb information takes away a very large part of the attractiveness of full closed looping.

And entering *imprecise* carb info could easy be inferior to not doing *any* carb inputs = to letting the *UAM mode* of oref(1) figure out further carbs that probably come to be absorbed in the next minutes, judging from the pattern of the calculated past *carb deviations* (see [section 4.5](https://openaps.readthedocs.io/en/latest/docs/While%20You%20Wait%20For%20Gear/Understand-determine-basal.html#understanding-the-basic-logic-written-version) and <https://openaps.readthedocs.io/en/latest/docs/While%20You%20Wait%20For%20Gear/Understand-determine-basal.html#understanding-the-basic-logic-written-version>).

PS: Because that is so, also loopers who do carb inputs get the UAM predictions besides their other predictions, and their algo makes a judgement (every 5 minutes) as to what the best calculation might be for where glucose, underlying „real“ carb absorption, and estimated carb deviation are headed.

292 13.5 Machine Learning

293

294 Involving machine learning (“artificial intelligence”) could help both in the learning/tuning phase, but
295 also in fine adjustments in daily utilization.

296

297 The study “The artificial pancreas and meal control” that was already referenced ([section 1.2](#) , or
298 [https://github.com/bernie4375/HCL-Meal-Mgt.-ISF-and-IC-settings/blob/HCL-.settings-main-repo-](https://github.com/bernie4375/HCL-Meal-Mgt.-ISF-and-IC-settings/blob/HCL-.settings-main-repo-(pdf))
299 [\(pdf\)](#)) discusses *on page 80* the application of machine learning in some predictions of postprandial
300 glucose response (IEEE Control Systems Magazine, ResearchGate: The Artificial Pancreas and
301 Meal Control. A. El Fathi et al, IEEE Control Systems Magazine Feb.2018 p.67-85.).

302 So there is already a body of data and evidence. To which extent it lends itself to UAM remains to
303 be researched. For this, lots of data would have to be captured from UAM loopers, and I fear many
304 more data would be required than what could easily be captured in Clarity® or even the AAPS
305 logfiles.

306

307 In the DIY universe, a prototype solution (to some aspects) was already developed for AIMI
308 ([section 13.3.3](#)).

309

310 We might see industry come up with a 1st generation solution that will probably be geared to folks
311 with miserable HbA1c and poor carb counting/meal handling, to offer a safe gradual improvement.

312

313 A top performing entirely self-learning system might be impossible to design:

314 For instance, if today you do something entirely different from yesterday (don't we all want this
315 freedom – even need it? Think about the fasting day following a feasting day...) there are
316 two problems:

- 317 • Such systems rely on information from the preceding day, or an average of several preced-
318 ing days
- 319 • The user does not know/learn much about how the system works, what it is calibrated for
320 today, how she/he might intelligently change something for the specific different situation
321 coming up. This seems like the opposite of the FCL solutions we discussed, for instance
322 self-defined Automations, combined with profile switches for to-be-expected temporary sen-
323 sitivity shifts (see [section 13.1](#) , or the more sophisticated options presented for exercise in
324 [section 6.1.3](#) and [6.2](#)).

325

326 13.6 Dual Hormone Systems

327

328 Off-topic: Besides using a glucagon analogue as a second independent hormone, there is also
329 research on just adding very small amounts into the (one) insulin cartridge. Then it seems to work
like an additive to make Lyumjev even faster acting, which would be a good thing notably for FCL.

Brief discussion and further reference on this topic see in “Insulins, DIA...pdf” at:

<https://github.com/bernie4375/HCL-Meal-Mgt.-ISF-and-IC-settings>

This is not our topic in this section, though.

Many see a **dual hormone full closed loop** as the ultimate system.

The beauty of this concept would be that the second (cartridge in the) pump could influence the glucose curve via giving glucagon or an analogue, thus overcoming the strongest limitation our current systems have:

Taking basal away (zero-temping) is only a severely limited course of action against impending hypoglycemia, and therefore, to keep things safe at the back-end of each meal, fighting glucose highs is more limited than we would like to see.

In conclusion, the glucagon component **not only helps stay out of hypos. It enables a more aggressive treatment for preventing, or reducing, high glucose values, as well.**

While insulin and carbs have complex activity curves stretching over hours, glucagon has a window of physiological activity starting 5-10 minutes after administration, and lasting only 30-40 minutes. Compared to insulin and carbs, that makes it a better component for rapid corrections (without a lengthy “tail” of action).

As glucagon does not per se introduce more calories, but stimulates glucose release from the liver, there should at least be no concern about gaining body weight from eventual roller-coasters the dual loop might send us into. Actually there could be a nice side benefit of helping in body weight control. Also, activity/sports management could become as easy as the meal management became in the UAM step into full closed looping.

It will be interesting to see for which application(s) the dual loop will be developed and launched; as part of a full closed loop with top performance, or as part of even only a hybrid closed loop for problem patients?

It remains to be seen how well such systems work in day-to-day circumstances. And whether “real people” will be able to handle all the involved technology, and use it in ways that truly could justify the substantial extra cost.

The author currently is not really looking forward to become loaded with even more technology, and quite happy with an aggressively tuned full UAM closed loop (...and an occasional nice post-dinner, or before/during-exercise snack).

However, the dual hormone path holds enough promise to learn more about it, and to test it some time in the near future.

369 13.7 Co-Medications that might facilitate FCL

370

371 Although the hype around GLP.1 and GIP receptor agonists (Tirzepatide, Mounjaro® and
372 others) is about weight loss, and in most markets type1 diabetics are not even eligible to
373 get a prescription, there is the interesting hypothesis, that **slowed digestion makes the**
374 **job significantly easier for a full closed loop**. More importantly, it could lower the
375 hurdles for fine-tuning, and open up an avenue to use FCL to a broad population of T1Ds.

376

377 Aside from the FCL aspect, the tendency towards improved %TIR in the T1D population-
378 was demonstrated in a few smaller studies already, e.g.:

379 **Efficacy and Safety of Tirzepatide in Adults With Type 1 Diabetes:** A Proof of Concept Obser-
380 vational Study (J Diabetes Sci Technol 2025 Mar;19(2):292-296; PMID: **38317405** PMCID:
381 [PMC11571402](#) DOI: [10.1177/19322968231223991](#)) [Halis Kaan Akturk](#)¹, [Fran Dong](#)¹, [Janet K](#)
382 [Snell-Bergeon](#)¹, [Kagan Ege Karakus](#)¹, [Viral N Shah](#)²

383 **Background:** Tirzepatide is approved by the United States Food and Drug Administration (FDA)
384 for the management of type 2 diabetes. The efficacy and safety of this drug have not been studied in
385 people with type 1 diabetes (T1D).

386 **Methods:** In this single-center, retrospective, observational study, hemoglobin A1c (HbA1c),
387 weight, body mass index (BMI), and continuous glucose monitoring (CGM) data were collected
388 from electronic health records of adults with T1D at initiation of tirzepatide and at subsequent clinic
389 visits over 8 months. Primary outcomes were reduction in HbA1c and percent change in body
390 weight and secondary outcomes were change in CGM metrics and BMI over 8 months from base-
391 line.

392 **Results:** The mean ($\pm$ SD) age of the 26 adults (54% female) with T1D was 42 ± 8 years with a
393 mean BMI of 36.7 ± 5.3 kg/m². There was significant reduction in HbA1c by 0.45% at 3 months
394 and 0.59% at 8 months, and a significant reduction in body weight by 3.4%, 10.5%, and 10.1% at 3,
395 6, and 8 months after starting tirzepatide. Time in target range (TIR = 70-180 mg/dL) and time in
396 tight target range (TITR = 70-140 mg/dL) increased (+12.6%, $P = .002$; +10.7%, $P = .0016$, respec-
397 tively) and time above range (TAR >180 mg/dL) decreased (-12.6%, $P = .002$) at 3 months, and
398 these changes were sustained over 8 months. The drug was relatively safe and well tolerated with
399 only 2 patients discontinuing the medication.

400 **Conclusions:** Tirzepatide significantly reduced HbA1c and body weight in adults with T1D.

401

402 Follow the discussion regarding utility in our diabetes management e.g. here:

- 403
- Discord: <https://discord.gg/YvrHYmdamc>
 - Diabettec: <https://www.diabettech.com/glp1-ra/glp1-ras-and-type-1-diabetes-a-retrospective-study-of-incretin-based-therapy-use-with-open-source-aid/>
- 405

406 **Call for a case study submission if someone was able to test this.** *Note: The author had*
407 *proposed a study but could not get a prescription to actually do it. See Discord.*

408

409

410 **Final Remark**

411

412 This is an exciting time to be part of the open source T1D community. Anyone is welcome to
413 contribute ideas, help develop software or instructions how to use.

414 Carefully weigh for yourself what may be your entry point for eventually surmounting the initial
415 hurdles, and **JUST EAT happily ever after**.

416